ENGINEERING MECHANICS: COMPREHENSIVE MASTER TEXTBOOK

1. Principles of Statics

Equilibrium of Rigid Bodies

A body is in equilibrium if the sum of all forces and moments acting on it is zero:

$$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M = 0$$

Free Body Diagrams (FBDs) are essential for visualizing all external forces, including reaction forces and applied loads.

2. Kinematics and Kinetics

Newton's Second Law

The motion of a particle is governed by $F = ma$. Kinematics deals with the geometric aspects of motion (displacement, velocity, acceleration) without considering the forces that cause them.

3. Work and Energy

Work-Energy Theorem

The work done by the resultant force on a particle is equal to the change in its kinetic energy.

$$W = \Delta K.E. = 0.5 * m * (v_2^2 - v_1^2)$$

4. Long-Answer Problem Bank

Problem 1 (Statics): A beam of length 10m is supported at both ends. A point load of 50kN is applied at 4m from the left end. Find the reaction forces at the supports.

> **Detailed Solution:**
> 1. Let R1 be the left support and R2 be the right support.
> 2. Σ Fy = 0: R1 + R2 = 50kN.
> 3. Σ M(left support) = 0: (50kN * 4m) - (R2 * 10m) = 0.
> 4. 200 = 10 * R2 => R2 = 20kN.
> 5. R1 = 50 - 20 = 30kN.

Problem 2 (Dynamics): A block of mass 10kg is pulled by a force of 50N on a frictionless surface. Find the acceleration and distance covered in 5 seconds starting from rest.

> **Detailed Solution:**
> 1. Acceleration (a) = F/m = 50N / 10kg = 5 m/s^2.
> 2. Distance (s) = ut + $0.5at^2$.
> 3. s = 0 + 0.5 * 5 * $(5)^2$ = 0.5 * 5 * 25 = 62.5 meters.

5. Friction

Laws of Friction

Frictional force (F) is proportional to the normal force (N): $F = \mu N$. The coefficient of friction (μ) depends on the nature of the surfaces in contact.